Yingrui Hou

Simulation & Model Validation | Digital Twin Applications

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience connecting measurements to simulation, calibrating models and diagnosing discrepancies. Built Python/C++/Geant4 workflows; learned TensorFlow through code adaptation and simulation-based verification, then migrated computation to a shared GPU workstation. Brings research coordination and technical communication to simulation engineering and digital twin opportunities.

Core skills

- Simulation: Geant4/C++, detector geometry, interaction stepping, response logic, event-level outputs and measured response maps.
- Validation: simulation-to-measurement alignment, deep-learning calibration, image-recognition model checks, bias correction and uncertainty analysis.
- Signals & models: time-profile decomposition, likelihood fitting, statistical inference, CatBoost/gradient boosting and data-driven model correction.
- Computing: Python, C/C++, ROOT/RooFit, NumPy, pandas; TensorFlow/GPU code adaptation and verification; Linux and shared GPU workstation setup.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Built and adapted Geant4/C++ detector simulations, implementing geometry, stepping and analysis outputs to study optical and detector response.
- Connected beam-test measurements and position-dependent response maps to simulated energy deposits, evaluating performance against measured rather than ideal response.
- Developed time-resolved signal analysis and ROOT/C++ correction workflows; checked reconstructed energy before and after correction under the studied conditions.
- Used real data to evaluate and optimise the ECAL clustering model; investigated reconstruction discrepancies to guide calibration and software follow-up.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Measurement-informed detector model

Integrated measured beam-test response maps into Geant4 energy-deposition simulation and compared measured and idealised responses. Non-uniformity contributed below 1% to the constant term in the first GRAiNITA prototype study. Transfer: model calibration and validation for digital twins.

Signal decomposition and correction

Extended simulated scintillation timing and fitted signal-component fractions to correct reconstructed energy. A representative rotation-based PSD study reduced a resolution term by about 2×; material comparisons exposed separation limits. Results apply to the studied configuration.

Real-data validation of learned models

Calibrated and validated a deep-learning flavour tagger; used real data to check and optimise an image-recognition model for ECAL clustering. Connected model-performance findings to calibration and reconstruction investigations.

Learning TensorFlow and delivering GPU computation

Learned TensorFlow from inherited code; personally ported amplitude models, added efficiency, resolution and bias effects, and validated calculations against mature C++ code at identical sampling points. Resolved discrepancies by aligning variable types; checked simulated observables against real data with KS tests. Set up a shared GPU workstation in half a day, then achieved a GPU run in another half-day. Contributed to testing a collaborator-developed fitter.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Learning, delivery & collaboration

- Learning by implementation: extended unfamiliar TensorFlow code with new research models and checked it on generated data; applied CatBoost and Geant4 in other projects.
- Delivery: agreed component interfaces and variable definitions before coding; reassessed priorities and timelines with collaborators as data processing exposed new issues.
- Collaboration: GitLab development and mutual merge-request review; identified a fitter sign error through mathematical checks. Weekly planning and rapid technical discussions; research presentations at GDR-InF 2022 and FCC/DRD France 2025.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Measurement-informed detector model (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)
- Signal decomposition and correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-5)
- Real-data validation of learned models (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Learning TensorFlow and delivering GPU computation (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)